

Course: Big Data Analytics

Course Code: 303105361

Prerequisite: Database Management System, SQL

Rationale : Big data analytics is the often-complex process of examining big data to uncover information such as hidden patterns, correlations, market trends and customer preferences that can help organizations make informed business decisions.

Teaching and Examination Scheme										
Teaching Scheme					Examination Scheme					
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
3	0	0	0	3	20	20	-	60	-	100

SEE - Semester End Examination, T - Theory, P - Practical

Course Content				
Sr.	Topics	W	T	
1	Introduction: What is in Store?, Classification of Digital Data: Structured, Semi-Structured & Unstructured, Evolution of Big Data, Definition of Big Data – Volume, Velocity, Variety, Challenges of Big Data, Why Big Data?, Traditional Business Intelligence (BI) versus Big Data, Industry examples of Big Data, What is Big Data Analytics?, Data Science.	20	9	
2	NoSQL Data Management: Introduction to NoSQL, Types of NoSQL, Why NoSQL?, Advantages of NoSQL, Comparison of SQL, NoSQL and NewSQL, Aggregates, Key-value and document data models, Graph databases, Map-reduce, Partitioning and combining.	20	9	
3	Basics of Hadoop: What is Hadoop?, Brief History of Hadoop, Why Hadoop?, RDBMS versus Hadoop, Hadoop Components, High Level Architecture of Hadoop, Key Advantages & Features of Hadoop, Data format, Hadoop Distributed File System (HDFS), Processing Data with Hadoop. Map Reduce Interface: Overview of Map Reduce, Map-Reduce workflows, Anatomy of Map-Reduce job run, Shuffle and sort, Task execution, Input formats, Output formats.	40	18	
4	Hadoop Related Tools: Overview of HBase, Pig introduction, Pig data model, Hive, Data types and file formats, HiveQL data definition, HiveQL data manipulation, HiveQL queries, Pig Latin Overview, Pig versus Hive, Using JSON, Overview of Cassandra, Jasper Reports.	20	9	
		100	45	

W - Weightage (%), T - Teaching hours

Reference Books

1.	Hadoop: The Definitive Guide, Third Edition, O'Reilly (TextBook) By Tom White
2.	Understanding Big Data, McGraw Hill, 2012 By Chris Eaton, Dirk Deroos et al.
3.	Hadoop Operations, O'Reilly, 2012 By Eric Sammer
4.	Big Data Analytics with R and Hadoop, SPD By Vignesh Prajapati
5.	Big Data and Analytics, Wiley India By Seema Acharya and Subhashini C
6.	Programming Hive, O'Reilly By E. Capriolo, D. Wampler, and J. Rutherglen
7.	MongoDB in Action, Dreamtech Press By Kyle Banker, Peter Bakkum, Shaun Verch
8.	HBase: The Definitive Guide, O'Reilly By Lars George

Course Outcome

After Learning the Course the students shall be able to:
1. Understand the Big Data flow.
2. Solve problems using MapReduce technique.
3. Implement single-node/multinode Hadoop cluster.
4. Differentiate between conventional SQL query language and NoSQL.
5. Apply the various technologies and tools associated with Big Data such as HDFS, MapReduce, Pig, Hive, MongoDB.